

Larry Huang

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IT FDN 130 A Sp 25: Foundations Of Databases & SQL Programming

Assignment #6

GitHub: <https://github.com/AAAcokies/DB-Foundations>

Disclaimer: this paper was partially written with ChatGPT's 4o model, with heavy manual modifications to fit my purposes and learned nomenclatures.

Views, Functions, Stored Procedures

Introduction

When working with databases, it's common to run into situations where you need to organize or reuse complex queries. That's where things like **views**, **functions**, and **stored procedures** come into play. Each serves a slightly different purpose, but they all help make databases work more efficiently, more organized, and easier to manage. Below are 2 common examples.

When Would You Use a SQL View?

A SQL view is used when you want to simplify complex queries by saving them as virtual tables. Instead of writing a long **SELECT** query every time you need to pull the same type of information, you can just query the view. Views are great when you want to hide complicated joins or calculations from users, enforce a specific way to look at the data, or create a consistent reporting structure. They don't store data themselves; they simply "save" a query that runs when you call it. They can also hide data you don't want users to be able to see. In short, views make things easier when you want reusable, readable access to slices of your database.

What Are the Differences and Similarities Between a View, Function, and Stored Procedure?

While views, functions, and stored procedures all help manage repetitive or complex tasks, they each have their own role. A **view** acts like a virtual table; you select from it just like a real table, but it's basically a saved query. A **function** is a bit different: it can take input, run code (like a calculation or lookup), and returns a single value or a table. You can use functions inside regular SQL statements (for example, in a **SELECT** or **WHERE** clause). A **stored procedure**, meanwhile,

is more like a mini-program inside the database. Procedures can take parameters, perform complex operations like inserting, updating, or deleting data, and even include control logic like **IF** statements and loops. Functions are generally more restricted and are meant to return something without changing the database, while procedures are more flexible and can change data.

Summary

Views, functions, and stored procedures are all about making SQL work smarter, not harder. Views simplify what users see; functions give calculated results or reusable logic inside queries; stored procedures handle bigger, multi-step operations. Knowing when and how to use each one makes database work a lot cleaner and way more efficient.